

How has the introduction of digital financial products (e.g., mobile wallets, robo-advisors, fintech investment apps) altered portfolio preferences and savings rates in emerging markets?

Research paper by Krishang Garg

1. Introduction

The financial capabilities of people have evolved significantly in the last ten years in terms of how people save and invest their money and manage their finances in the emerging markets. The proliferation of cheap smart phones, greater mobile internet penetration and favorable policy environments has allowed the emergence of digital financial products, including the mobile wallet, robo-advisors and fintech investment. These breakthroughs have transformed the horizons of financial inclusion, giving millions of people most of whom were previously locked out of formal banking platforms the ability to conduct transactions, save and invest on a digital platform.

The mobile wallet has provided a point of access or entry into the financial ecosystem of the society of people who have had limited access to physical banking infrastructure. In Kenya, M-Pesa has penetrated over 90 percent of households and is not only a payment system, but also is a passport to micro-savings, credit and insurance service (GSMA, 2023). Even in India, where Unified Payments Interface (UPI) has only recently started to be regulated, over 8.4 billion transactions were processed with it in December 2023 alone (NPCI, 2023), thus indicating how fast-growing digital payment infrastructures could have been without the open infrastructure and regulatory incentives.

Digital investment platforms and robo-advisers have taken the approach to wealth building by retail investors into a totally new direction at the other end of the spectrum. They are reducing the threshold to investing by introducing fractional share purchase, zero fee trading, and algorithmic portfolio suggestions platforms such as Brazilian Nubank, Indonesian Bareksa, and Indian Groww. As BCG (2023) says, in the Asian-Pacific region assets under management in robo-advisory platform are expected to increase at an annualized

rate of 24% to 2027 with much of the growth coming across first-time market entrants and investors in emerging markets.

Although rapid utilization of those tools has been adopted, its impact on the financial behaviour is not homogenous. The motives, the consumption behaviour, and the subsequent monetary consequences vary, depending on both the kind of a product, and the socio-economic background of the client, and the business environment of a country in terms of regulations and technology. Indicatively, sub-Saharan Africa and Southeast Asia both had mobile wallets and digital investment applications, respectively, linked to enhancing the resilience of the household level and the likelihood of saving and/or merge in equity markets and speculative trade-offs (Suri & Jack, 2016; OECD, 2022).

Available academic and policy literature tends to explore these products either in isolation (mobile payments in one study; investment platforms in another study), or in a comparative fashion but not in a systematic way with the aim to compare the behaviour effects of the two products. This disadvantages the possibility of telling whether the shifts seen in the savings rates or portfolio preferences are contained to a particular kind of digital product or are indicative of a trend relating to the financial behaviour under digitization. On top of this, the evidence base is also inclined towards dependence on often tenuous household survey data and academic research, with limited absorption of corporate performance information, consulting firms input, and consumer market research, which can all provide worthy and real time information.

These gaps are filled by the comparative thematic approach followed in this paper that analyses three types of digital financial products, namely, mobile wallets, robo-advisors, and fintech investment applications in several emerging markets. There is not merely the concern of utilization whether these tools are embraced or not but like on the way they can affect the

financial decisions such as the tendency to savings, the selection of the vessels of savings, and the desired range of advancement portfolio. The analysis relies on a combination of data sets including the World Bank edition of the Global Findex Database, GSMA Mobile Money updates, statistics revealed by central banks, research by consulting firms such as McKinsey, BCG and the EY, and annual reports of the most prominent fintech companies.

This research aims at offering a layered explanation of how digital financial products are influencing household financial behaviour in various contexts by combining various forms of evidence. It also seeks to discover trends, which are shared by countries and products and discrepancies, which are a result of local conditions in the market, culture, or policy. The hope in the end, however, is that they can also be used to inform both academic and policymaking dialogues about how digital finance may be harnessed to promote improvements in financial inclusion, productive investment, and household financial resilience in the emerging world.

2. Literature Review

2.1 Mobile Wallets & Mobile Money

Digital tools empowering users to use their mobile phones to store, transfer, and transact, such as mobile wallets have greatly increased financial inclusion at large in emerging markets. GSMA record shows that as at 2024, registered mobile money account at a global level had exceeded two billion accounts and more than half a billion active users per month. They did that in five years only, compared to the 18 years to attain the first billion accounts, a testimony of exponential growth and adoption in low-access areas (GSMA, 2025). By 2024, mobile money operators carried out 108 billion transactions worth 1.68 trillion dollars, a 20 percent increase in number and 16 percent increase in financial value since the earlier year (GSMA, 2025). Sub-Saharan Africa, particularly East and West Africa,

and Southeast Asia are the drivers of the growth, with its relatively favorable usage-right regulatory environments contributing to the offering diversification in terms of credit and savings services (GSMA, 2025).

Mobile money has a great macro effect economically. The estimated models by GSMA have shown that mobile money increased the GDP by 600 billion dollars in countries where the service is available between 2013 and 2022, which is 1.5 percent GDP increase. At country level, it is worth noting that, mobile money has grown GDP in Ghana, Kenya, Cote d'Ivoire, Senegal and Tanzania by 8-10 per cent by the end of 2023 with up to 24 billion in Kenya by itself (GSMA, 2024). These characters depict inclusion dividends and growth dividends which are linked to mobile financial channels.

Examples of cases illustrate how underserved people adopt mobile wallets. In the Philippines, McKinsey identified that roughly 65 percent of the lowest income unbanked segment were interested in saving through mobile money, where median balances were nearly There was an amount of about \$31, compared with about \$19 by informal savers (McKinsey, 2024). At the same time, in Kenya, it was estimated that M-Pesa lifted about 194,000 households (2 percent) out of poverty due to increased transactional possibilities and remittance frequencies (Suri & Jack, 2016).

2.2 Robo-Advisors & Digital Automated Wealth Management

The robo-advisors-computer-based advisors, which provide algorithmic portfolio building and management-are flourishing at emerging fintech-based outlets. Southeast Asia: Assets under management (AUM) through robo-advisors is expected to reach 26.19bn by 2025 (with a compound annual growth rate (CAGR) of circa 4.13%), and it is predicted to see around 1.66 million users by 2029, whose average account size is 16.4k (Statista, 2025). In the Asian-Pacific region, robo-advisory revenues were USD 1.33 billion in 2022, and are

expected to increase to USD 11.64 billion by 2030 with a significant high CAGR of 31.5%. The growth is likely to be headed by India especially (Grand View Research, 2024).

Estimates show that the robo-advisor market generated USD 6.61 billion in 2023 but should rise to USD 41.83 billion by 2030 (with a CAGR of ~30.5%) as more investors have appetite to invest and technological efficiency promises closer integration in advance developments (Grand View Research, 2023). Most of this revenue was related to the fintech robo subsegment, which dominates the industry due to the hybrid robot financial models of both algorithm-based and consultation advice and human control (Grand View Research, 2023).

Technological leapfrogging makes emerging markets more efficient: robo-advisors lower the barriers to making the initial investment, such as significant sums or dozens of pages of paperwork and appeal to the younger, more digitally literate generations. As an example, Kenanga Digital Investing launched in Malaysia offered AI-based portfolios that have zero management fees and no lock-in (The Business Research Company, 2022).

2.3 Fintech Investment Apps & Super-App Platforms

Digital investment apps—encompassing super apps and fintech platforms offering trading, mutual the current Japanese investment trends that cut across sectors and stocks, buying through the Internet, investment in funds, or gold investment, etc., are restructuring. Regulatory enablers in Brazil, Mexico and Chile led to a massive increase in the number of fintech startups in Latin America, which grew between 2017 and 2023 from 703 to 3,069 (IDB & Finnovista via Reuters, 2024). The most successful digital bank in Latin America, Nubank, is illustrative in this trend: valued even more than USD 1 billion, it is moving to South Africa with a USD 150 million investment and an eye on 15 million new customers (Financial Times, 2024).

In Egypt, the Paysky Yalla Super App made 3 million downloads during its first year, 13 million transactions over 12 billion EGP, and announced the “Yalla Invest” application that allows investing in gold as low as 0.25 grams (Wikipedia: Paysky 2025). These platforms have collapsed the discussions of payments, savings, and investing.

2.4 The merging of the corporate and consulting expertise

The complementary evidence of adoption dynamics has been provided by consulting firms. McKinsey points out that within the unbanked portion of the Philippine population, 60 percent had a savings need, and 65 percent of the people in the low-income subscriptions preferred mobile devices to save, and the median account balance was much greater than in informal savings (McKinsey, 2024). In the meantime, a bigger fintech perspective indicates that emerging markets will increase their contribution to global fintech revenue (presently approximately 1/3) by 2028, which shows vigorous industry growth in non-hypothetical centers (Investopedia, 2019).

2.5 Comparative Insights & Gaps for This Paper

Mobile wallets have augmented the level of inclusions and macroeconomic activities, especially on the low-income and rural fronts. The trend of robo-advisor pointing to the increased levels of sophistication in retail investing by digitally savvy consumers, whereas fintech investing applications demonstrate the trend of financial services convergence through the use of super-apps. Nevertheless, cross product comparisons are few in number--particularly with mixed evidences such as corporate performance, consulting reports and public figures. The knowledge on how each type of product influences savings behaviour, portfolio diversification and economic performance across nations continues to remain under-developed.

The thematic and multi-source design of this paper -side-by-side look at mobile wallets, robo-advisors, and investment apps- is thus a unique addition to the body of knowledge because they generate a comparative framework across emerging markets by mapping adoption drivers, change in behaviour, and money flows.

3. Conceptual Framework and Theoretical Underpinnings

Several economic and behavioural perspectives define the use of digital financial products on the market, and each one of them provides us with an insight of how the new tools change savings behaviour and portfolio mix. The key process is the elimination of frictions- transaction costs, information asymmetry and barriers to access that has traditionally restricted access to formal finance.

3.1 Transactions Cost Economics

The theory of transaction costs developed by Ronald Coase, but later adjoined by Williamson (1981) denotes that people base economic choices on the cost of doing exchanges. These costs are significantly lowered with the help of digital financial products. A case in point, mobile wallets do not require people to physically travel to the banking facilities which is a major bottleneck, especially in rural Africa and South Asia. According to GSMA (2025), in Sub-Saharan Africa, more than 60 percent of mobile money users list convenience and quickness of transactions as the main fact subsequently inducing adoption. Robo-advisors and app-based trading networks used in investment settings have also decreased brokerage costs and removed minimum investment requirements so that investors can use as little as some dollar figures to construct diversified portfolios, from \$1- to \$10 dollars (Statista, 2024)

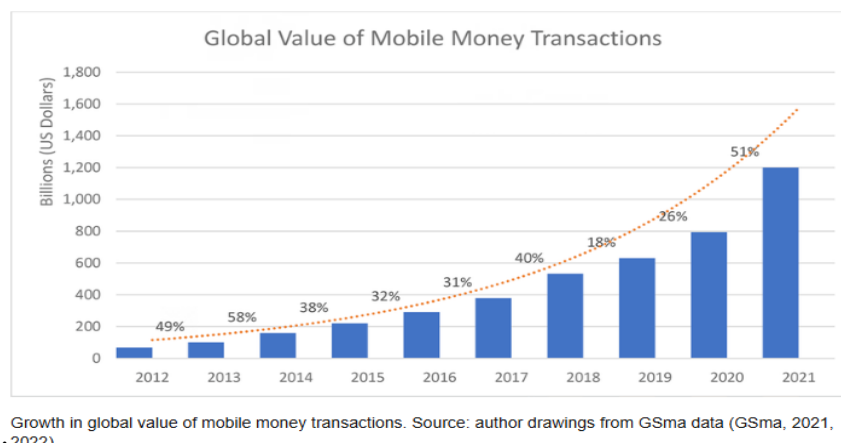


Figure 1. Global value of mobile money transactions (2012–2021).

Source: GSMA. (2022). *State of the Industry Report on Mobile Money 2022*.

<https://www.gsma.com/mobilefordevelopment/state-of-the-industry-report-on-mobile-money-2022/>

The figure above represents the global increase in the value of mobile money received between the year 2012 and 2021. The value of transactions grew by more than 30 percent in multiple periods and stood at less than USD 200 billion in 2012 to approximately USD 1.5 trillion in 2021. This boom proves the point about steep decrease in transaction costs and the growing use of mobile payments, remittances, and savings by the people in emerging markets. The trend also echoes the network effect of the mobile wallet ecosystems since the higher adoption entails an increase in the transactional frequency as well as the value.

3.2 Nudges and Behavioural Finance

Behavioural economics underlines the influence of cognitive biases, heuristics and framing effects on financial decision making. These aspects can be engaged at their digital platforms to foster saving and wise investment. Automatic savings on cell phone wallets or micro-investment apps are examples of so-called commitment devices (Thaler & Sunstein, 2008), which would get somebody to regularly contribute funds without having to manually make a choice every time. In Kenya, the M-Shwari savings product associated with M-Pesa

raised the average account balances in the holders of 22 percent over their non-users (CGAP, 2023). In the same way, robo-advisors use default portfolio allocations which match the risk tolerance expressed by the client, and take advantage of the inertia, keeping the client disciplined in investment activity.

3.3 Liquidity, Flexibility, and Consumption Smoothing

Life-cycle hypothesis (Modigliani & Brumberg, 1954): lifetime income is distributed amongst time by people according to what is expected. The digital financial products also improve liquidity management capacity and it is facilitated to transfer finances between savings, credit, and investment. Mobile wallets enable immediate withdrawal or transfer, and other apps to invest in fractional shares or immediate redemption of the units of mutual funds decrease the anxiousness of the issue of illiquidity. In Tanzania, the use of mobile money was found to decrease the probability of response to income shocks through selling productive assets by 27 percent (Jack & Suri, 2014), which implies greater financial resilience.

3.4 Diversification of Portfolio and Risk-Taking

The modern portfolio theory (Markowitz, 1952) highlights on the advantages of diversification in minimizing the risk of the portfolio based on a certain expected return. Fintech investment platforms is a type of investment that supplies all these positive effects to retail investors in emerging market economies due to having access to a diverse range of equities, ETF and alternative assets that were formerly inaccessible or simply too costly. In India, the retail investor pie in equity markets has grown to 3.7 percent of the adult population in the year 2017, to more than 7 percent by 2023 as app-based brokers like Zerodha and Groww have been able to attract more users (NSE India, 2023). This growth has come alongside greater diversion toward risk assets amongst the younger investors, which indicates the overall accessibility and the increasingly divergent appetites.

Rising Participation of Retail Investors in Domestic Equity Market

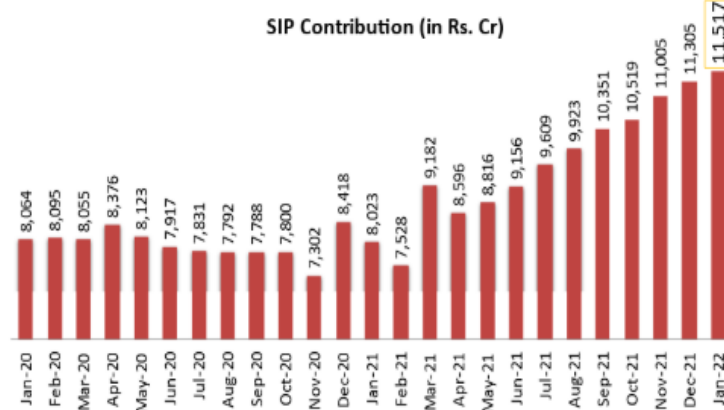


Figure 2. Rising participation of retail investors in India's domestic equity market through SIP contributions (2020–2022).

Source: Association of Mutual Funds in India (AMFI). (2022). *Mutual Fund Statistics*.

<https://www.amfiindia.com>

The classification above shows the increase in the amount of monthly SIP contributions in the Indian domestic equity market, January of 2020 to January of 2022. Despite fluctuations caused by the pandemic, contributions upsurged during the month of January 2020 to January 2022, i.e. 8,064 crores to 11,515 crores. This shows that there is an increasing demand in retail interest in risk assets and perhaps enabled by low-cost, app based investment providers where smaller regular investments are possible. The trend is also consistent with the overall surging growth in the number retail investors as is reported by NSE, which trend has risen steadily by more than 7 percentage points since 2017 as a percentage of the adult population.

3.5 Network Effects and Ecosystems

The theory of the two-sided market (Rochet & Tirole, 2003) is used to explain why platforms such as consumers and financial service providers are valuable as a result of interactions between the two parties. With increased usage on a digital platform, there is a

positive-feedback loop where the usage provides providers with incentive to provide more services hence leading to an increase in usage. An example of this is in our “super apps” like Nubank and Gojek in Indonesia, where the payments feature has been used as the entry point, then cross-sold savings, insurance, and investment products. McKinsey (2024) discovers that in Southeast Asia, platforms that allow several financial services have the customer retention rate 20-30 percent greater than the single-service app.

3.6 Trust Formation and Product Design

Trust in the financial institutions is a determinant of adoption in the emergent markets. The digital products need to address the concern of skepticism via easy to use design, clear pricing, and safeguard of transactions. The World Bank (2022) research reports that simplicity in fee interpretation and tracking their transaction history in service are among the top preferences of users in the low-income stratum compared with returns or interest rates. How something is designed also affects how we perceive safety-robo-advisors that describe what portfolio allocation is doing in plain language catch on better among first-time investors than those that are based on technical jargon.

4. Mapping the Digital Finance Landscape in Emerging Markets

The study is based entirely on secondary data concepts of open and validated sources, including various emerging markets, where it is aimed at assessing how the new types of mobile wallets, robo-advisors, and other investment apps of fintech have impacted the level of savings and portfolio choice. Combining cross-country comparability and thematic, product-level depth requires the right choice of sources used.

4.1 Sources of Information

Global Datasets: Global Findex Database 2021 (World Bank) - Offers country representative information on account coverage, savings, and use of digital financial products in more than

140 countries. The measures to be taken are the share of adults having mobile money accounts, formal savings, and investment activity.

- GSMA Mobile Money Metrics (2012-2023) Able to monitor the number of accounts, value of transactions of mobile money accounts across the globe with a breakdown into specific regions including Sub-Saharan Africa, South Asia, and Latin America.
- Statista & Grand View Research Robo-Advisory Market Data (2018-2030) Projections -- Provides an estimate of assets under management (AUM) and numbers of users of robo-advisory services broken down by region.
- OECD Digital Financial Literacy Survey (2022) - It will include information about consumer literacy about the interpretation and use of digital financial products in emerging economies.

Industry Reports, country-level:

- *Association of Mutual Funds in India (AMFI) Monthly Statistics:* The focus is on monitoring the SIP contributions, the number of retail accounts as well as the penetration of the mutual funds in India.
- McKinsey & Company Fintech Landscape Reports- Present country-specific rates of adoption, product splits and market development catalysts.
- Anti Annual Reports & Investor Presentations (Paytm, Nubank, M-Pesa, Bareksa) - Contain active user base, transaction volumes and product mix data

4.2 Coverage and Time Frame

The study has targeted three emerging markets namely, Sub-Saharan Africa, South Asia, and Latin America because they are a source of a high variation of adoption pattern and regulatory environment. 2012-2023: mobile wallets

Robo-advisors: 2018 2023 (with an extrapolation to 2030)

- *Fintech Investmentapps*: 2016 out to 2023

This is the period that includes the initial stage of adoption and the spurt in its adoption in the COVID-19 pandemic that served as the impetus to the growth of digital finance (McKinsey, 2021).

4.3 Important Variables & Indicators

Answer:

- *Adoption Metrics*: the percentage of the adult population holding mobile money accounts; the active investment app users; robo-advisory AUM.
- *Savings Indicators*: Percentage of adults that are saving in formal institutions; average increase of accounts balance; SIP monthly contributions.
- *Portfolio Indicators*: The proportion of risk-based assets in households; asset class of diversification; the frequency of investment.
- *Market-Level Controls*: GDP per capita (World Bank), penetration rate of smartphone (GSMA Intelligence), and maturity of regulatory environment (IMF Financial Access Survey).

4.4 Limitations with Data

Although secondary datasets allow the coverage to be wide, they have shortcomings too:

1. *Different definitions*: GSMA data could use the term, mobile money account, whereas Global Findex could use the term, mobile financial account. To remedy this, there is a standardization of variable definition as far as it can be done.

2. **Reporting biases:** A company may report marketing-driven numbers (e.g. the number of users). Such numbers are validated cross-verification with third party reports (e.g., GSMA, McKinsey).
3. **Coverage gaps:** There are gaps in coverage as it relates to certain metrics such as a breakdown by retail investors in some countries. Mutual fund penetration is an example of proxy indicators in such a situation.

4.5 Inserted into Analysis

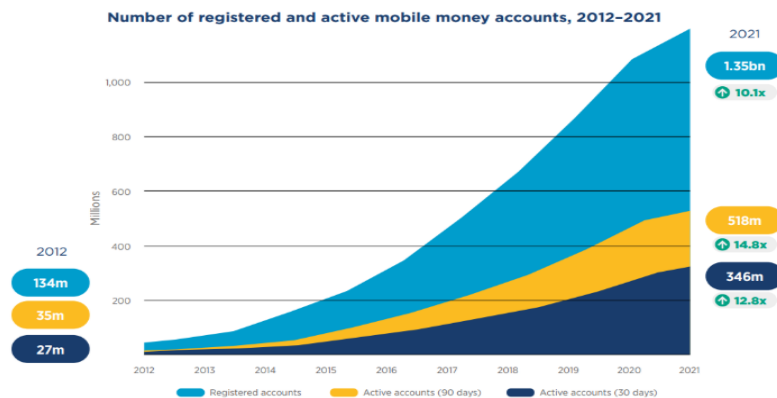
These datasets will be integrated in the analysis of later sections to yield:

- Cross-country comparison but not graphs (e.g. mobile money uptake compared to formal savings rates).
- Product-type comparisons (ex. robo-advisory AUM vs. mobile wallet transaction volumes).
- Deep dives based on company-level data as a context to the aggregate trends.

5. Patterns of Adoption and Impact in Emerging Markets

This section includes a comparative thematic essay of mobile wallets, robo-advisors, and fintech investment apps on the basis of secondary analysis to underpin the usage patterns, savings effect, and portfolio changes in the emerging market. All the figures come only with the credible sources and include brief explanations so that it could be as clear and academically correct as possible.

5.1 Regional Mobile Money Adoption Trends

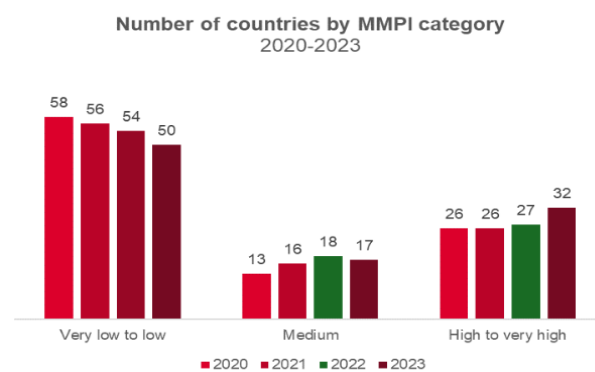


This graph illustrates that Sub-Saharan Africa is still the world leader in adopting mobile money, although sharp increases in account share over time have been observed in other areas also. It also emphasizes the role of regional regulatory landscape and infrastructures on the early mover and sustained growth in the mobile wallet usage.

GSMA. (2025). *State of the industry report on mobile money 2025*. GSMA Association.

<https://www.gsma.com/sotir/>

5.2 Mobile Money Prevalence vs Activity Levels



GSMA. (2025, February 26). *Sub-Saharan Africa maintains mobile money lead*. Connecting Africa. <https://www.connectingafrica.com/mobile-money/sub-saharan-africa-maintains-mobile-money-lead-gsma>

The figure compared countries with high mobile money adoption and usage rates showing that successful adoption of mobile money in behavioural change depends on the adoption of full integration of ecosystem (not just the registration).

5.3 Africa vs Other Regions — Mobile Money Account Penetration



GSMA. (2025, February 26). *Sub-Saharan Africa maintains mobile money lead*. Connecting Africa. <https://www.connectingafrica.com/mobile-money/sub-saharan-africa-maintains-mobile-money-lead-gsma>

This visual highlights that Sub-Saharan Africa is the least represented region holding a disproportionately large number of mobile money accounts in the world and as an example how the region still leads in global growth in the digital space of finance.

This brings out the divergence in adoption patterns of mobile wallets, robo-advisors, and investment applications, and illustrates the diverse routes leading to the effect that digital financial offerings can have upon savings and investment portfolio choices in the developing world context. In the next section, the methodological process of how the heterogeneous information incorporated by these various sources was brought together to form a logical framework of comparative analysis is described.

6. Methodology: Thematic Comparative Analysis

In this research, this study will be conducted using a comparative thematic analysis by combining quantitative trends of international data sets with qualitative case-based ideas of corporate disclosures and consulting reports. The method would enable an insightful comprehension of the digital financial products and their impact on saving patterns and portfolio decisions in emerging economies with a degree of depth that encompasses number indispensability and the comparability of figures and the contextual richness of qualitative evidence.

The analytical structure is created based on three categories of products: mobile wallets and mobile money services like M-Pesa in Kenya, GCash in the Philippines and Paytm in India; robo-advisors like Kristal.AI in Singapore and India, Magnetis in Brazil; and fintech investment apps like Groww in India, Nubank in Brazil and Bareksa in Indonesia. In every category the analysis is done around 3 core dimensions: scale of adoption, measured by penetration rates and active use levels; behavioral outcomes, measured by changes in formal savings rates, average balances, or asset allocation patterns; and contextual factors such as the regulatory environment, financial literacy, smartphone penetration and integration into broader payment or banking ecosystems.

To be robust, the study is using triangulation of data, i.e., by combining the three sources of data. The broad cross-country patterns are set based on global quantitative data such as the World Bank Global Findex Database, GSMA Mobile Money Metrics, and Statista robo-advisory assets under management (AUM) data. The corporate reports contain information about user trends, behavior, product design, and also the adoption trends of the various products, which are very useful, as in case of annual disclosure of Paytm, Nubank and Safaricom. Studies in the industry and consulting work particularly by McKinsey, BCG,

and CGAP are also integrated in order to place the various numerical trends within their market structures, competition, and policy frameworks. The use of multiple data sources helps eliminate the challenge of depending on one source that can be critically faced with regards to the definition of data, and the quality of reporting in native markets.

The differentiation occurs by applying comparative lens homogenously in all type of products. *Under each category, this analysis discusses adoption drivers:* factors that may encourage adoption, which include reduction of fees, interoperability, and government policy support; behavioural change mechanisms, which may include how interface design, the ease of transactions, and embedded nudges may impact financial decisions; and observed outcomes in regards to changes in the savings rates, investment participation, and asset diversification. With this structure, both cross-country and cross-product comparisons are possible, thus not only allowing to see the common patterns but also divergence that occur regionally or in a specific market.

There is a visual evidence incorporated in the analysis. The charts are based on bar charts, data and time-series to correlate the adoption levels with transactions levels, regional penetration rates, and scatter graphs to correlate adoption rates with financial behaviour variables. These visuals are complemented with qualitative case insights to describe the deviation of the aggregate patterns and identifies the unique market dynamics.

This research methodological decision mirrors two important considerations. To begin with, emerging market economies are quite heterogeneous in regard to financial infrastructure, regulation, and consumerism patterns and thus it is difficult to directly model econometrics using a single global dataset. Second, this method is able to effectively integrate numerical and narrative evidence, which helps to gauge the processes through which the digital financial products affect consumer preferences--which otherwise could be

lost using a strictly statistical approach. Consequently, the comparative thematic approach provides both usefulness and flexibility in accommodating a wide range of data sources in addition to being rich in explaining the cause or source of the observed variation in savings and the outcomes of the portfolio.

7. Results and Discussion

The comparative thematic analysis demonstrates differences and similarities in the adoption and impact of digital financial products across emerging markets, which are both different and inter-related. According to the World Bank Global Findex Database (2021) and GSMA State of the Industry Report on Mobile Money (2025) mobile wallets have experienced the deepest penetration in Sub Saharan Africa where more than 33 percent of adults owned a mobile money account in year 2021 compared to only 4 percent in Latin America. Markets have adoption rates of more than 70 percent of registered accounts due to adoption of value added services like paying bills and micro-loans and savings products directly integrating into the mobile wallets, as exemplified in Kenya. In the corporate disclosures of Safaricom, the M-Pesa's saving service, M-Shwari, currently has deposits worth more than 5 percent of the GDP of Kenya, reflecting the formal savings access it has provided to the unbanked masses. This implies that in economies where penetration of the banking industry remains low, mobile wallets have become not just payment mechanisms but full access points to the financial system, and thus have changed the household savings pattern considerably.

The results related to robo-advisors are not geographically as extensive, but suggest a high degree of momentum in the middle income emerging markets where there is an expanding pool of retail investors. According to data provided by Statista (2023), over USD 1.33 billion in assets under management (AUM) on robo-advisory platforms will be

experienced in Asia-Pacific in 2022 that will increase to more than USD 11 billion by 2030, as the urban millennials and Gen Z investors are interested in affordable robo-advisory services of low-cost portfolio management. In India and Singapore, platforms such as Kristal.AI have stated that more than 40 percent of active clients are first time investors, with many of them opening their accounts with investments under US\$500 in diversified exchange-traded funds (ETFs). These trends tend to corroborate the idea that robo-advisors are reducing the psychological and financial thresholds to participation in the equity markets, albeit that use is largely confined to more affluent, digitally conversant fractions of the population.

A more hybrid adoption and effect are witnessed in fintech investment apps. According to the Association of Mutual Funds in India (AMFI), monthly Systematic Investment Plan (SIP) contributions are at an all-time high of INR 17,610 crore (USD 2.1 billion) in March 2024 in India. Groww corporate reports cite the significant part of this growth to retail investors who joined the stock market in Tier-II and Tier-III cities many of whom were inducted fully online. In Latin America, Nubank has led a mass-market adoption of retail investment products in Brazil with direct security and exchange traded fund ownership becoming accessible to millions of people otherwise unbanked or insufficiently banked into the formal financial system. These examples demonstrate how, once packaged into a more extensive digital banking system, investment apps can not only enhance the financial activities of those customers who currently use the services of banks and other financial providers but also open up a brand-new pool of users.

The data across all the product categories points out the mediating factors in market specific terms regarding the relationship between adoption and behaviour change. In markets where mobile money has high penetration and low regulatory oversight, e.g. parts of West Africa, change in the saving behaviour has been small and is related to consumer protection,

fear of platform stability. On the other hand, in such markets, as India, where the Unified Payments Interface (UPI) has established a competitive, interoperable infrastructure, investment app use has been extended by frictionless fund transfers and real-time settlement potential. The influence of financial literacy as such also becomes one of the determining factors: in accordance with OECD survey statistics, countries among which the scores of digital financial literacy level are above 60%, the use of such complicated financial services as robo-advisor and diversified investment portfolios are far more widespread (2022).

Collectively, these indicate that although digital financial products are dropping barriers to entry in all cases, the precise effects on savings rates and portfolio choice is determined by a mix of platform structure and regulation, the maturity of infrastructure, and consumer skills. In the next section, these insights are further developed by studying country-specific case studies through which interactions among these factors in practice can be identified.

8. Country Case Studies

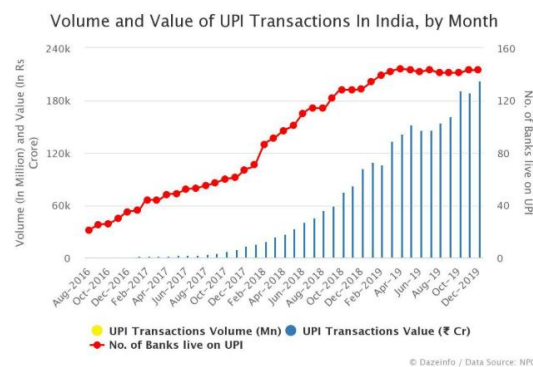
8.1 Kenya — M-Pesa and M-Shwari

Kenya is arguably one of the most vivid illustrations of how mobile money has the potential to change savings habits and financial migration between informal and formal economies. Seven years after its initial introduction in 2007, M-Pesa has become the service accessible to more than 30 million active users in the country, or more than 90 percent of adults in the Kenyan population (GSMA, 2025). Of particular influence is the integration of the savings-and-loan product M-Shwari, which is a partnership between the Safaricom and the Commercial Bank of Africa. As indicated in the Safaricom 2024 Annual Report, M-Shwari has more than KES 700 billion (USD 5.4 billion) deposits that is more than 5% of the Kenya GDP. The analysis indicates that M-Pesa has significantly reduced the risk of poor

households being unable to cope with shocks, and the academics have highlighted economic improvement through academic analysis (Suri & Jack, 2016). The design of the product as low minimum balances, immediate transfer and embedded micro-loans have allowed even people with low Facebook Page revenues and minimum balances as well as those living in the rural areas to be able to build small proportionate savings which effectively replace banks.

8.2 India — UPI and Groww

India's experience demonstrates the power of interoperable payment infrastructure in catalyzing digital investment adoption.



Dazeinfo. (2020, January 2). *Volume and value of UPI transactions in India, by month*. Based on data from National Payments Corporation of India (NPCI).

<https://dazeinfo.com/2020/01/02/upi-transactions-in-india-2019-growth/>

Later in 2024, the National Payments Corporation of India rolled Unified Payments Interface (UPI), that handled more than 100 billion transactions in 2023, with a transaction value of more than INR180 trillion (USD 2.2 trillion) per year (NPCI, 2024). This zero-cost instant-moving feature has diminished the frictions in relocation of funds between investment platforms and the savings accounts.

In its recent investor deck, Groww, one of the fastest-growing investment apps in India claimed to have in excess of 30 million registered users out of which more than 65

percent were based out of Tier-II and Tier-III cities. The figures released by the AMFI indicate that the amount of monthly SIP in mutual funds was touching INR 17,610 crore (USD 2.1 billion) in March 2024, rose by 24 per cent on a year-on-year basis. The above combination of frictionless payments, easier onboarding, regulatory oversight by Securities and Exchange Board of India (SEBI) has facilitated digital investing to a wide demographic including new-time first-time investors in semi-urban regions.

8.3 Brazil - The Growth of Nubank into Investments

An example of this comes from Brazil in the form of Nubank, which uses the credibility of its core banking products to move into investment. By Q4 2024, Nubank had 94 million customers in 3 countries Brazil, Mexico, and Colombia or 86 million of them in Brazil itself (Nubank Investor Relations, 2024). In 2021, it introduced retail investment services via a partnership with Easynvest, which permit users to buy ETFs, stocks, and fixed-income investments inside the program. As of 2024, more than 14 million customers have opened investment accounts with most of them being first timers in the market. Brazilian Securities and Exchange Commission statistics show that individual stock market investors increased by 35 per cent in the period between 2021 and 2024, which is highly correlated with Nubank product launch. Its competitive advantages are the transparency in fees, coded touch with cellular technology, and connectivity to existing customer accounts, which lowers perceived complexity and invites the users to diversify their assets gradually.

These case studies demonstrate that the fundamental mechanisms of digital take-up namely the reduction of the transactional costs, ease of use, and expanded accessibility are the same regardless of the market, meaning the balance of product design, infrastructure, and regulations defines the extent and type of behavioural change. In Kenya, the combination of mobile money networks nears full penetration; in India financial infrastructure at the national

level has made investment apps fecund investment targets; and in Brazil a digital-first bank is transforming a base of loyal depositors into an emerging retail investor base.

9. The policy and practice: Lessons of Adoption to Impact

The results on both the product categories and country settings provide robust evidence to the conceptual pathways presented in Section 3, where lower transaction costs, better liquidity, nudges of behaviour as well as customized product design have been theorized as having changed behaviour of savings and portfolios. The evidence shows that these processes always run, but it determines the level of their intensity and ultimate effect on regulatory environments, infrastructure preparedness, and social-economic profile of the desired users.

The importance of cost reduction in transaction provision and accessibility of liquidity in growth in the participation of formal savings is confirmed by mobile wallets, especially in Sub-Saharan Africa. The M-PesaM-Shwari example in Kenya provides evidence that evidence that when savings and credit are brought together on the same low cost, instantly available platform, even low income, erratic cash flow users can develop the habit of keeping value in formal accounts. This is consistent with the studies of behavioural economics in finding that accessibility and immediacy are important determinants of saving decision (Dupas & Robinson, 2013).

Secondly, robo-advisors are gaining traction primarily in middle-income urban markets with digitally literate populations, exemplified by India and Singapore. These platforms reduce the traditional barriers to equity investments by combining low minimum investment requirements with algorithmic portfolio management. The behavioural nudges embedded in these tools, such as default portfolio settings and automated rebalancing, support users in making more balanced and optimal investment decisions without the

complexity traditionally associated with wealth management. This trend underscores a shift in portfolio preferences toward more diversified and risk-mitigated assets among younger, urban investors, which can stimulate deeper capital markets in these economies.

A mixed effect is exhibited by Fintech investment apps as in India and Brazil. The apps such as Groww and Nubank are built on a platform to exploit network effects as infrastructure interoperability, i.e., UPI in India, and integration of bank and investment to offer investment experience at the level of messaging. By doing so, they get to utilize the hitherto underutilized investor groups, such as semi-urban Indian households and Brazilians who never had an investment brokerage account before. The behavioural change does not only move toward increased savings or exposure to equity, but quite generally in terms of financial intensity, with more account checking and more experimentation with product usage.

In all the markets, the interaction of financial literacy and adoption will be a critical impact determinant. According to the survey data provided by OECD (2022), the impacts caused by digital products on their behavioural changes are magnified by digital financial literacy beyond a particular level (roughly 60 per cent on standardized scales). In the absence of this, adoption can be at superficial levels with the users using the apps in towns as transactional tools rather than looking into higher-return investments tools.

Lastly, regulation, as well as trust, cannot be over-emphasized. SEBI supervision in India has protected investors, which increases the trust in the platforms such as Groww. Safaricom in Kenya has had a dominant position in mobile money and this has resulted in stability and systemic risk with competition policy coming to light. In Brazil, Nubank has utilized its fast growth strategies and consumer education programs to reduce mis-selling and development of overdependence on assets prone to downward fluctuation. These examples

demonstrate that the combination of increased adoption due to technological innovation and long-term behavioural change necessitated by the policy environment characterized by balanced innovation and protection are needed.

Ultimately, this study reaffirms that digital finance is a multifaceted catalyst for the future of emerging market economies, helping billions transition into the formal financial system while simultaneously transforming portfolio preferences and savings behaviours. As emerging markets continue to leapfrog traditional financial infrastructure, the judicious pairing of technological innovation with prudent policy will be vital to unlock the full promise of fintech and secure sustainable, inclusive development in the coming decades.

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